

$$\theta|y \sim N(\hat{\theta}, [I(\hat{\theta})]^{-1})$$

$$(\theta - \hat{\theta})|y \sim N(0, [I(\hat{\theta})]^{-1})$$

Now consider

$$z = [I(\hat{\theta})]^{1/2}(\theta - \hat{\theta})$$

If $X \sim N(\mu, \Sigma)$ and A is a linear transformation, then

$$AX \sim N(A\mu, A\Sigma A^T)$$

mean: $[I(\hat{\theta})]^{1/2} \cdot 0 = 0$

variance: $[I(\hat{\theta})]^{1/2} [I(\hat{\theta})]^{-1} ([I(\hat{\theta})]^{1/2})^T = I$

symmetric

Hence $z|y \sim N(0, I)$

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$y \sim N(\theta, 1)$ uniform prior

$H_0: \theta = 0$

$H_A: \theta > 0$

Data: $y = 1$

Frequentist approach	Bayesian approach		
$z = \frac{1-0}{1} = 1$	$y=1$	likelihood	$p(y \theta)$
one-sided p-value of 0.16		prior	$p(\theta)$
two-sided p-value of 0.32	:		
BOTH "fail to reject"	Posterior probability that $\theta > 0$ is 89%		
p-value > 0.05 threshold	) updating a belief with uncertainty quantified.		